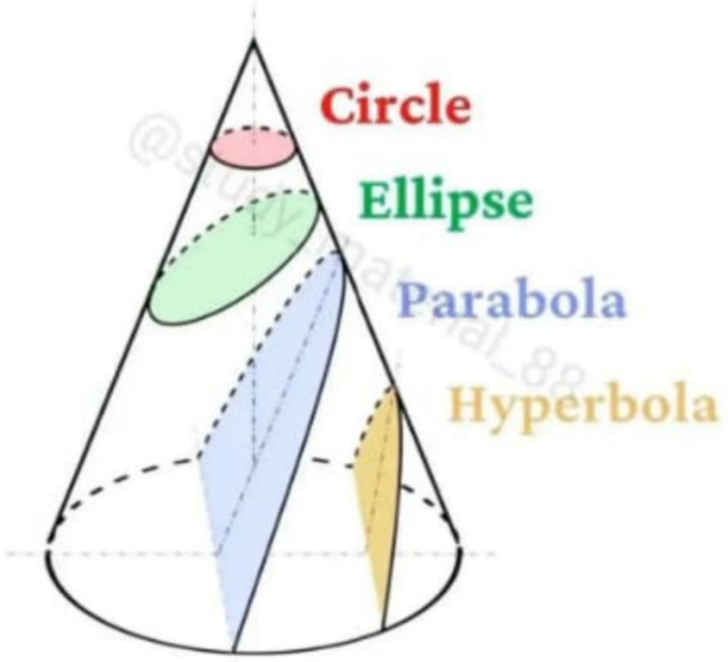
CONIC SECTIONS



- Ellipse
- Parabola
- Hyperbola

Ellipse:-

The best example of ellipse is the elliptical orbit of the earth around the sun.

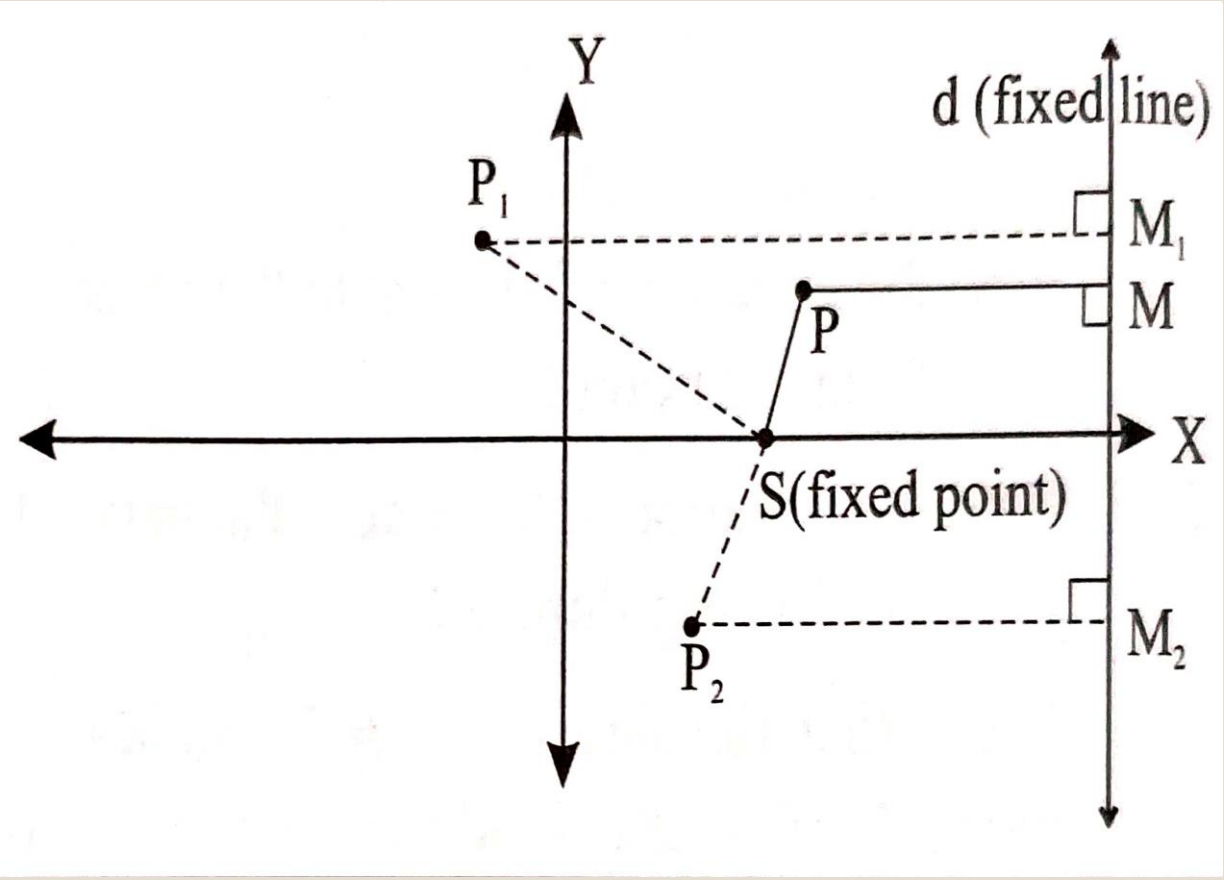
- Definition:-

1. An ellipse is the locus of the point in a plane which moves so that its distance from a fixed point bears a constant ratio $e(o<e<1)$ to its distance from a fixed line.
2. The fixed point is called the focus S and the fixed line is called the directrix d.

- Standard equation:-

$(x^2)/(a^2) + (y^2)/(b^2) = 1$, $a > b$

- Let S be the focus, d be the directrix and e be the eccentricity of an ellipse.
- Draw SZ perpendicular to directrix.
- Let A and A' divide the segment SZ internally as well as externally in the ratio e: 1.



Parabola:-

The u shape curve is called parabola.

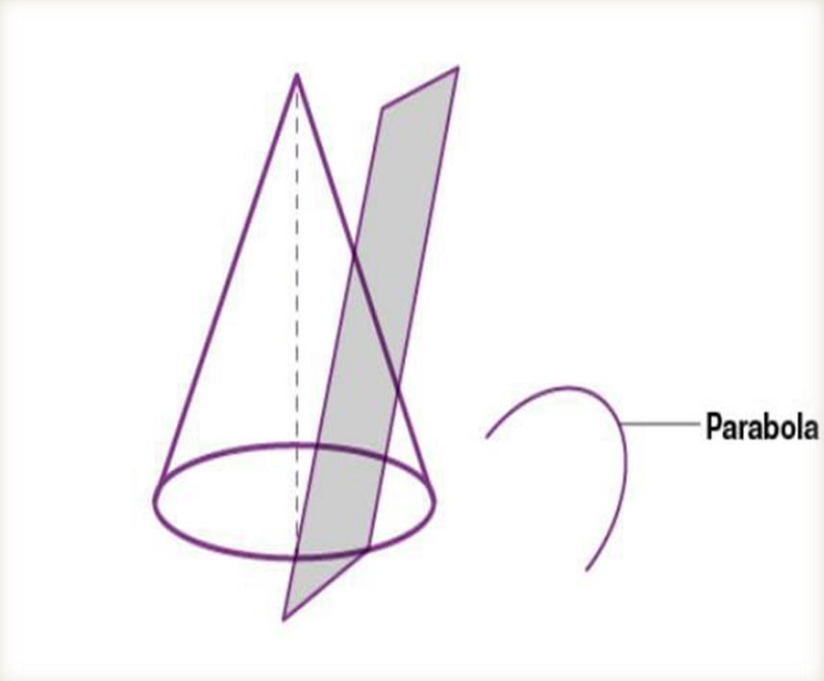
- Definition:-

A parabola is a locus of the point in plane equidistant from a fixed point and fixed line in that plane.

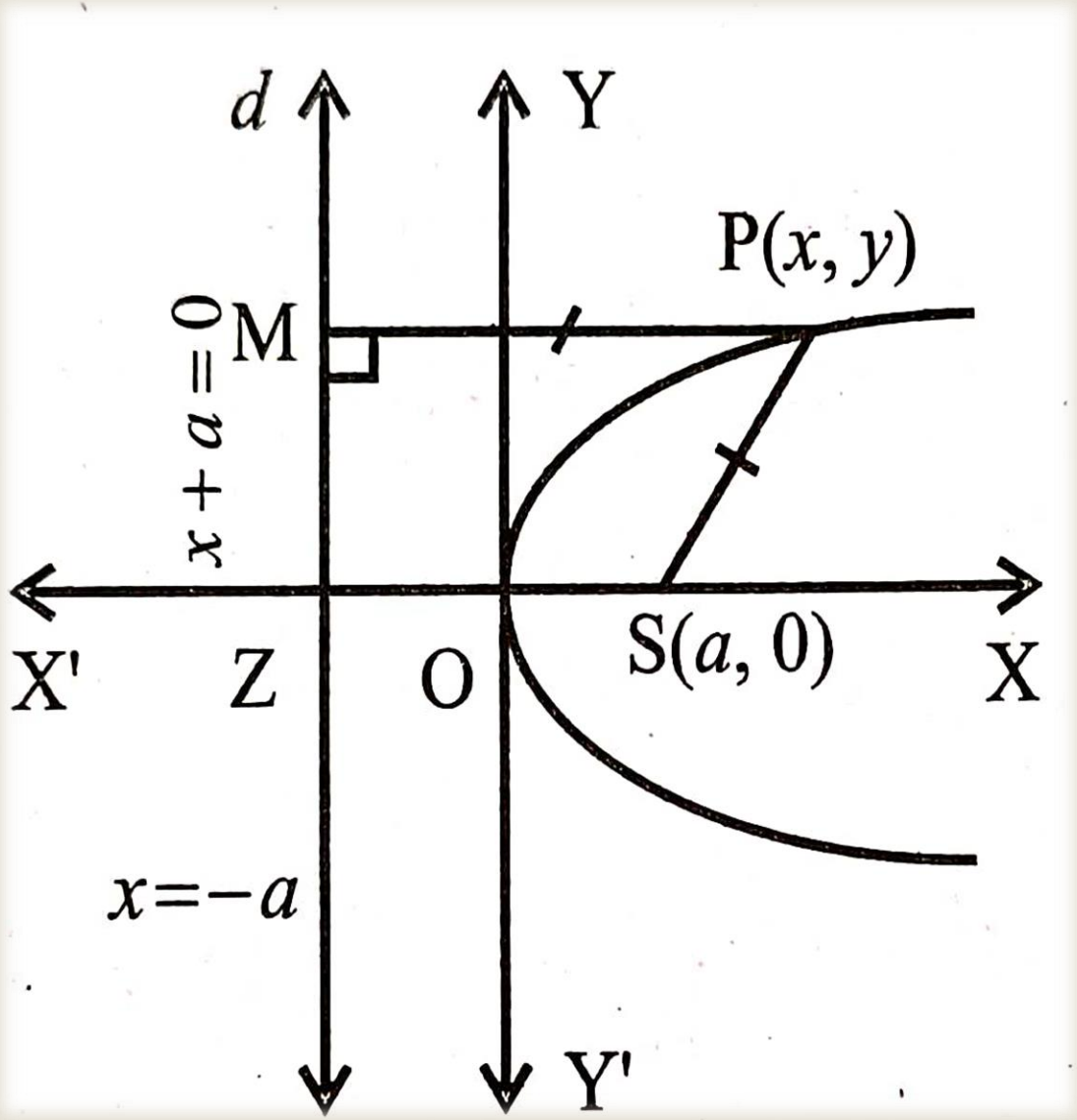
1. The fixed point is called focus
2. The fixed straight line is called the directrix.

- Standard equation of the parabola:-

$Y^2=4ax$



- Let S be the focus and d be the directrix of the parabola.
- The standard equation of the parabola is $Y^2=4ax$ $(a>0)$.
- Let SZ be perpendicular Y to the directrix.
- Bisect SZ at the point O.
- Midpoint O is on the parabola.
- Let Sz =2 a.a>0.



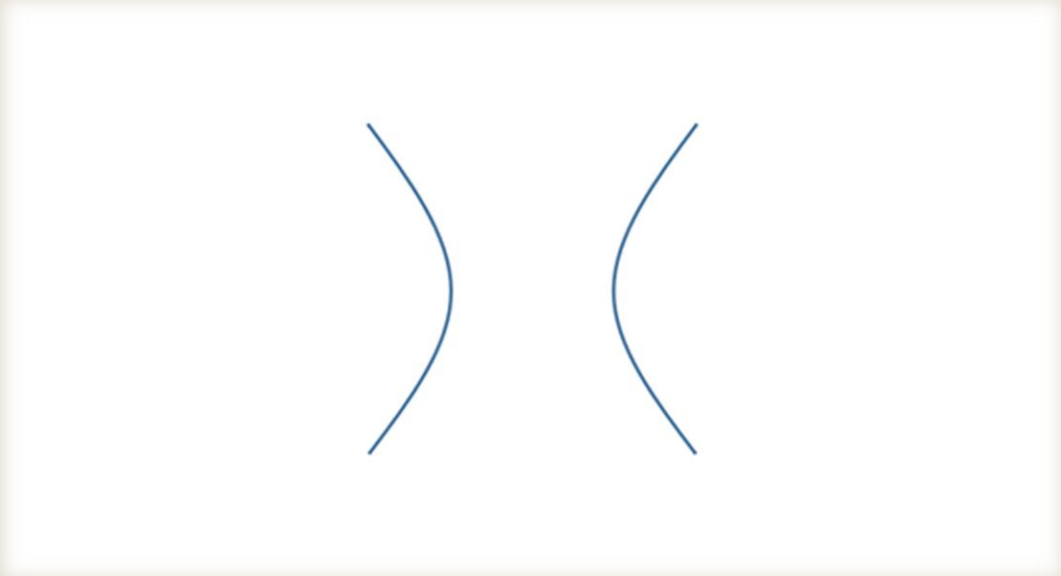
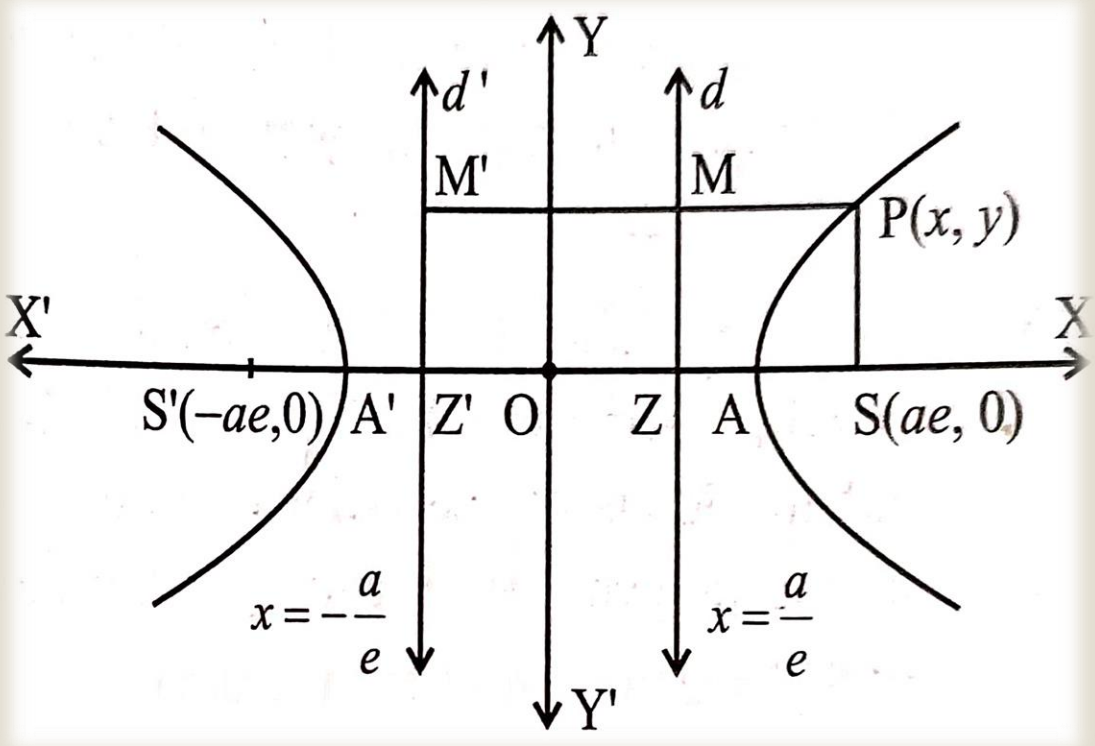
Hyperbola :-

- Definition:-

S is the focus and d is the directrix no containing the focus and P is the moving point PS PM then =e, where PM is the perpendicular on the directrix. e > 1 called eccentricity of the hyperbola.

- Standard equation:-

$(x^2/a^2) - (y^2/b^2) = 1$.



- Let S' be (- ae, 0) and d' be the line x = - a/e.
- For any point P on the hyperbola, PM' is perpendicular on d' then it can be verified that PS ' prime = ePM '.
- Thus for the hyperbola (a e,0) and (- a e, 0) are two foci and x = a/e corresponding two directrices.